

CACIC 2026 Supplementary Material

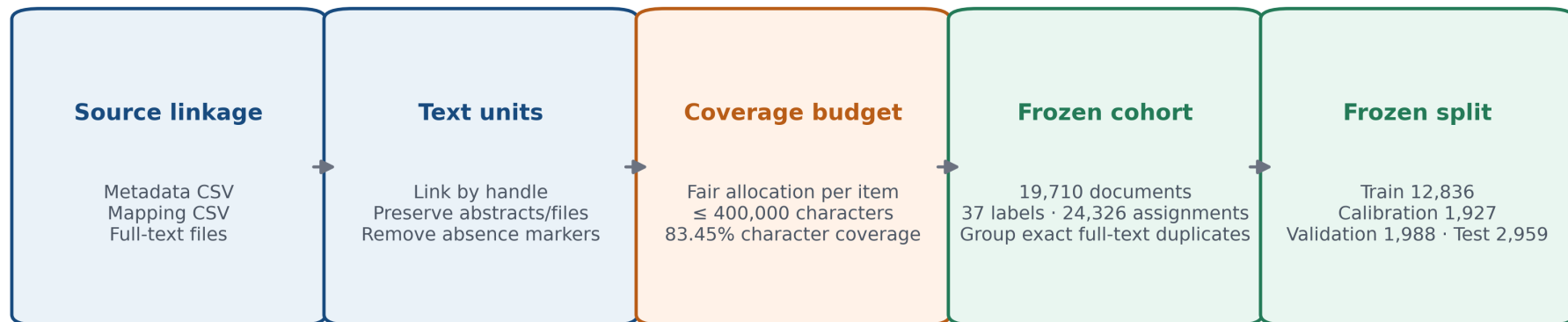
Beyond Embeddings: Language-Aware Sparse Representations and Fine-Tuned Multilingual Transformers

Aggregate reviewer evidence only. Repository texts, handles, item-level predictions, split manifests, model weights, credentials, and personal paths are excluded.

S1. Corpus and protocol

Property	Value
Documents	19710
Labels / assignments	37 / 24326
Abstract segments / full-text files	24411 / 20750
Items with multiple abstracts / full texts	4365 / 463
Represented full-text characters	83.45%
Train / calibration / validation / test	12,836 / 1,927 / 1,988 / 2,959

Dataset construction and leakage-controlled partitioning



All 37 labels occur in every partition; representations are fitted on train only.

Figure S1. Dataset construction and frozen partitions.

Calibrated multilingual long-document evaluation protocol

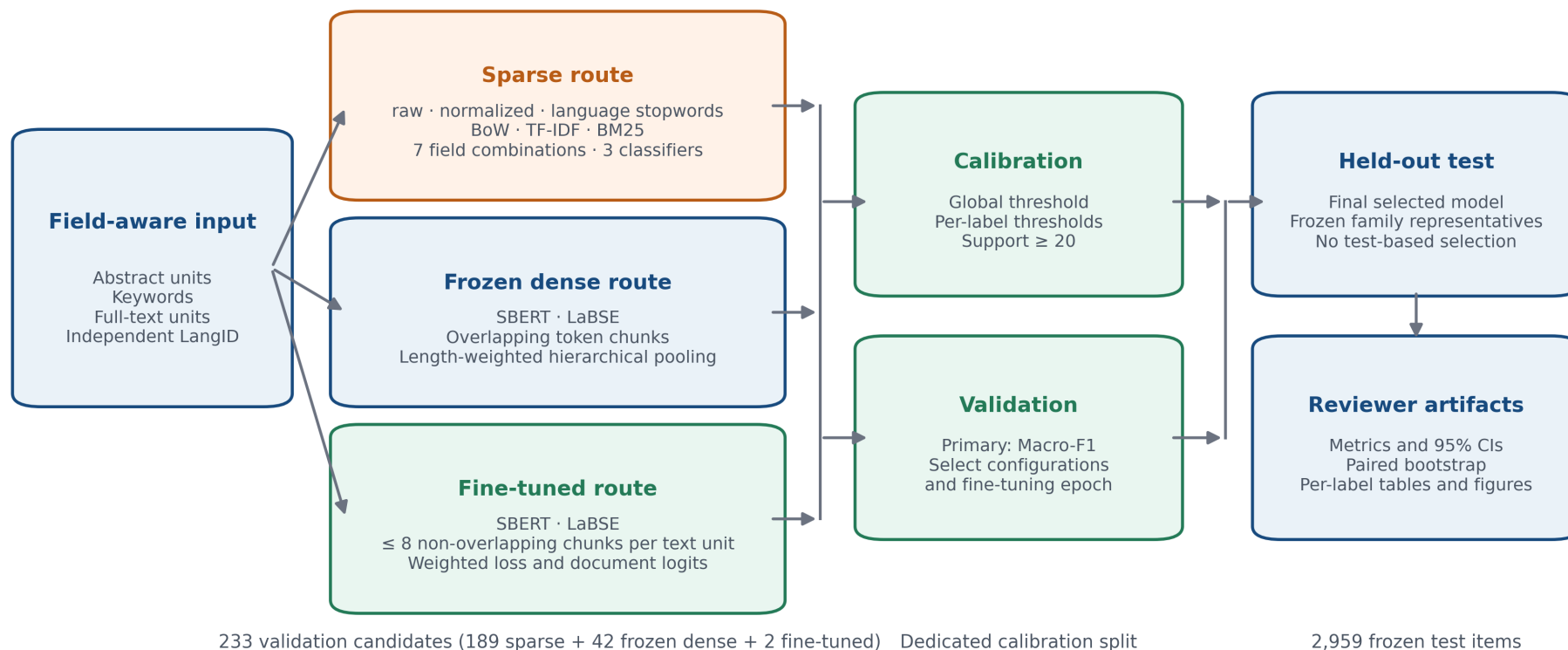


Figure S2. Experimental and selection protocol.

S2. Complete experiment overview

Rank	Configuration	Macro-F1	Micro-F1
1	bm25 / fulltext / language_stopwords / linear_svc	0.7483	0.7705
2	bm25 / abstract+keywords+fulltext / language_stopwords / logreg	0.7388	0.7774
3	tfidf / abstract+fulltext / language_stopwords / linear_svc	0.7357	0.7571
4	bm25 / fulltext / raw / linear_svc	0.7338	0.7480
5	bm25 / fulltext / normalized / linear_svc	0.7335	0.7484
6	tfidf / fulltext / language_stopwords / linear_svc	0.7329	0.7550
7	tfidf / keywords+fulltext / normalized / linear_svc	0.7316	0.7533

Rank	Configuration	Macro-F1	Micro-F1
8	tfidf / abstract+keywords+fulltext / language_stopwords / linear_svc	0.7315	0.7559
9	bm25 / abstract+fulltext / language_stopwords / logreg	0.7312	0.7752
10	tfidf / abstract+keywords+fulltext / raw / linear_svc	0.7296	0.7509
11	bm25 / keywords+fulltext / language_stopwords / logreg	0.7295	0.7704
12	tfidf / abstract+keywords+fulltext / normalized / linear_svc	0.7294	0.7512
13	tfidf / keywords+fulltext / raw / linear_svc	0.7294	0.7517
14	bm25 / abstract+keywords+fulltext / language_stopwords / linear_svc	0.7291	0.7612
15	tfidf / keywords+fulltext / language_stopwords / linear_svc	0.7291	0.7543
16	bm25 / fulltext / language_stopwords / logreg	0.7283	0.7718
17	bm25 / abstract+keywords+fulltext / language_stopwords / sgd	0.7276	0.7706
18	tfidf / fulltext / raw / linear_svc	0.7254	0.7490
19	bm25 / keywords+fulltext / language_stopwords / sgd	0.7251	0.7648
20	tfidf / fulltext / normalized / linear_svc	0.7249	0.7482

The complete 233-row grid is supplied as `paper_artifacts/runs/v3_main/results_validation_grid.csv`.

S3. Family-level held-out comparison

Family	Selected feature set	Validation Macro-F1	Test Macro-F1	Test Micro-F1
bm25	fulltext	0.7483	0.7457	0.7641
tfidf	abstract+fulltext	0.7357	0.7338	0.7529
labse_finetuned	abstract+keywords+fulltext	0.7148	0.7069	0.7279
bow	abstract+keywords+fulltext	0.6895	0.6782	0.7135
sbert_finetuned	abstract+keywords+fulltext	0.6663	0.6595	0.6746
sbert_frozen	keywords+fulltext	0.6278	0.6211	0.6397
labse_frozen	keywords+fulltext	0.6254	0.6143	0.6334

S4. Convergence audit

Configuration	Macro-F1	Micro-F1
bm25 / fulltext / language_stopwords / linear_svc_c05_tol5e4	0.7497	0.7732
bm25 / fulltext / language_stopwords / linear_svc_c05_tol1e4	0.7497	0.7732
bm25 / fulltext / language_stopwords / linear_svc_c1_tol5e4	0.7483	0.7705

Configuration	Macro-F1	Micro-F1
bm25 / fulltext / language_stopwords / linear_svc_c1_tol1e4	0.7483	0.7705
tfidf / fulltext / language_stopwords / linear_svc_c1_tol1e4	0.7329	0.7550
tfidf / fulltext / language_stopwords / linear_svc_c1_tol5e4	0.7329	0.7550
tfidf / fulltext / language_stopwords / linear_svc_c05_tol1e4	0.7307	0.7554
tfidf / fulltext / language_stopwords / linear_svc_c05_tol5e4	0.7307	0.7554

S5. Final held-out result

Metric	Estimate	95% CI
Macro-F1	0.7522	[0.7372, 0.7643]
Micro-F1	0.7685	[0.7567, 0.7797]
Subset accuracy	0.5863	—
Hamming loss	0.0164	—
Precision@3 / Recall@3	0.3810 / 0.9320	—

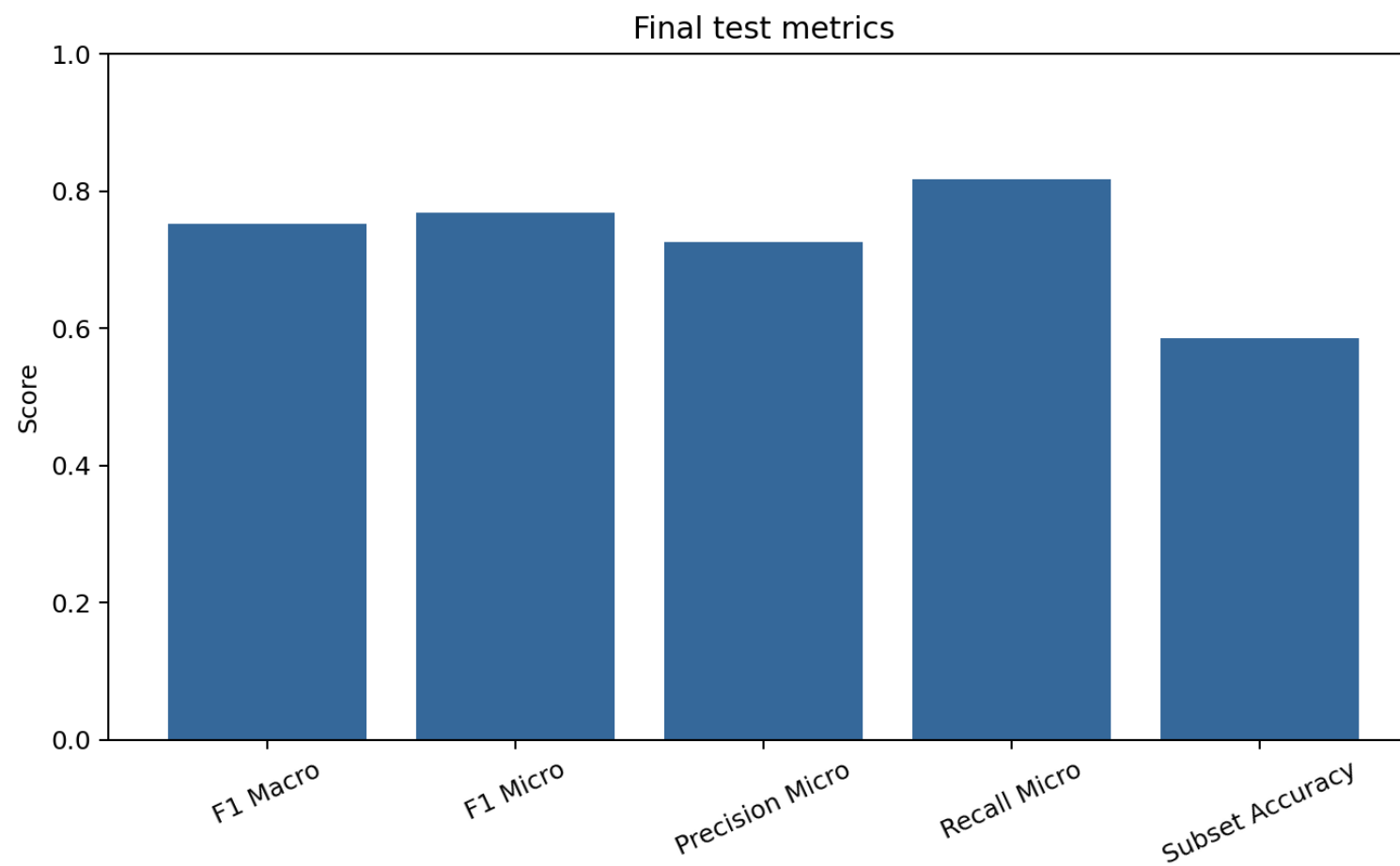


Figure S3. Final test metrics.

S6. Fine-tuned transformer validation

Model	Feature set	Selected epoch	Macro-F1	Micro-F1
labse_finetuned	abstract+keywords+fulltext	5	0.7148	0.7321
sbert_finetuned	abstract+keywords+fulltext	5	0.6663	0.6794

S7. Paired bootstrap comparisons

Candidate	Metric	Candidate – final BM25	95% CI	P(candidate > BM25)
tfidf	f1_macro	-0.0184	[-0.0299, -0.0068]	0.0000
tfidf	f1_micro	-0.0156	[-0.0243, -0.0072]	0.0010
labse_finetuned	f1_macro	-0.0453	[-0.0587, -0.0311]	0.0000
labse_finetuned	f1_micro	-0.0406	[-0.0508, -0.0303]	0.0000
bow	f1_macro	-0.0741	[-0.0896, -0.0590]	0.0000
bow	f1_micro	-0.0549	[-0.0662, -0.0441]	0.0000
sbert_finetuned	f1_macro	-0.0928	[-0.1070, -0.0779]	0.0000
sbert_finetuned	f1_micro	-0.0939	[-0.1055, -0.0822]	0.0000
sbert_frozen	f1_macro	-0.1311	[-0.1458, -0.1163]	0.0000
sbert_frozen	f1_micro	-0.1288	[-0.1398, -0.1175]	0.0000
labse_frozen	f1_macro	-0.1379	[-0.1528, -0.1233]	0.0000
labse_frozen	f1_micro	-0.1350	[-0.1469, -0.1226]	0.0000

S8. Per-subject results

Subject	Support	Precision	Recall	F1	Average precision
Odontología	66	0.9275	0.9697	0.9481	0.9792
Ciencias Informáticas	387	0.9623	0.9225	0.9420	0.9746
Educación Física	73	0.9211	0.9589	0.9396	0.9882
Relaciones Internacionales	115	0.8433	0.9826	0.9076	0.9674
Psicología	100	0.8774	0.9300	0.9029	0.9597
Ciencias Veterinarias	72	0.7753	0.9583	0.8571	0.9307
Letras	182	0.7907	0.9341	0.8564	0.9442

Subject	Support	Precision	Recall	F1	Average precision
Bibliotecología	37	0.8611	0.8378	0.8493	0.9084
Ciencias Agrarias	116	0.7754	0.9224	0.8425	0.8997
Bellas Artes	106	0.7388	0.9340	0.8250	0.8927
Ciencias Jurídicas	154	0.7245	0.9221	0.8114	0.9085
Ciencias Económicas	85	0.9672	0.6941	0.8082	0.9138
Ciencias Médicas	54	0.7385	0.8889	0.8067	0.8280
Sociología	165	0.7622	0.8545	0.8057	0.8616
Ingeniería	88	0.7207	0.9091	0.8040	0.8744
Arquitectura	77	0.8852	0.7013	0.7826	0.8961
Química	62	0.7105	0.8710	0.7826	0.8316
Filosofía	53	0.9714	0.6415	0.7727	0.8262
Geografía	35	1.0000	0.6286	0.7719	0.8621
Ciencias Naturales	186	0.7022	0.8495	0.7689	0.8314
Física	62	0.7143	0.8065	0.7576	0.8395
Ciencias Astronómicas	37	0.6471	0.8919	0.7500	0.8160
Historia	124	0.6522	0.8468	0.7368	0.8067
Antropología	59	0.6970	0.7797	0.7360	0.8191
Astronomía	35	1.0000	0.5714	0.7273	0.9160
Trabajo Social	46	0.7619	0.6957	0.7273	0.8109
Ciencias de la Educación	50	0.7447	0.7000	0.7216	0.7747
Zoología	33	0.6486	0.7273	0.6857	0.6669
Periodismo	88	0.6311	0.7386	0.6806	0.7561
Humanidades	227	0.5545	0.8062	0.6571	0.7860
Comunicación Social	93	0.6275	0.6882	0.6564	0.7186
Comunicación	103	0.5769	0.7282	0.6438	0.6717
Educación	185	0.5437	0.7405	0.6270	0.6475
Ciencias Exactas	126	0.7765	0.5238	0.6256	0.7233
Urbanismo	30	0.4340	0.7667	0.5542	0.6012
Ciencias Sociales	104	0.5316	0.4038	0.4590	0.4887
Biología	34	0.2152	0.5000	0.3009	0.2084

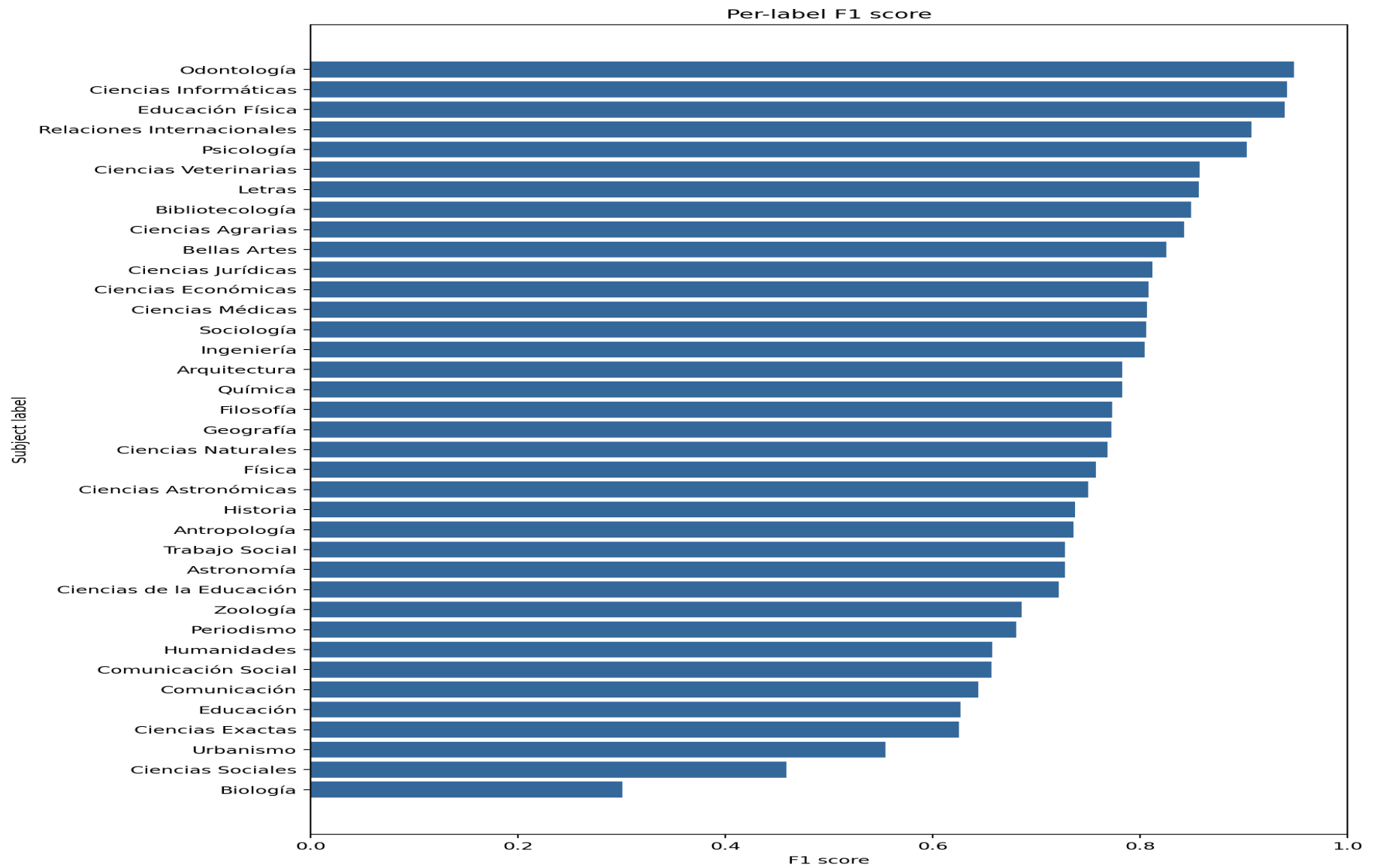


Figure S4. F1 by subject.



Figure S5. One-vs-rest confusion matrices.

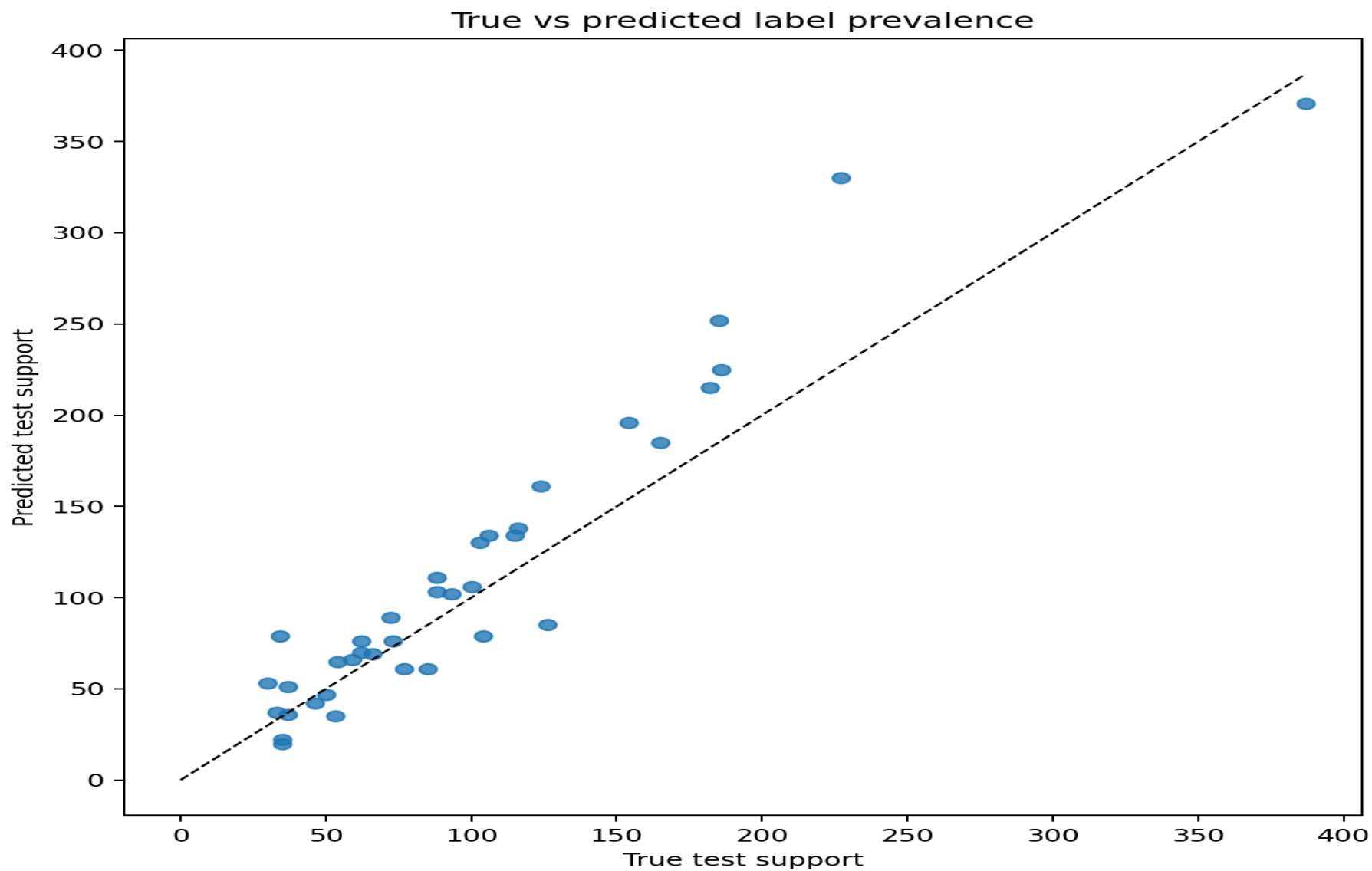


Figure S6. True versus predicted support.

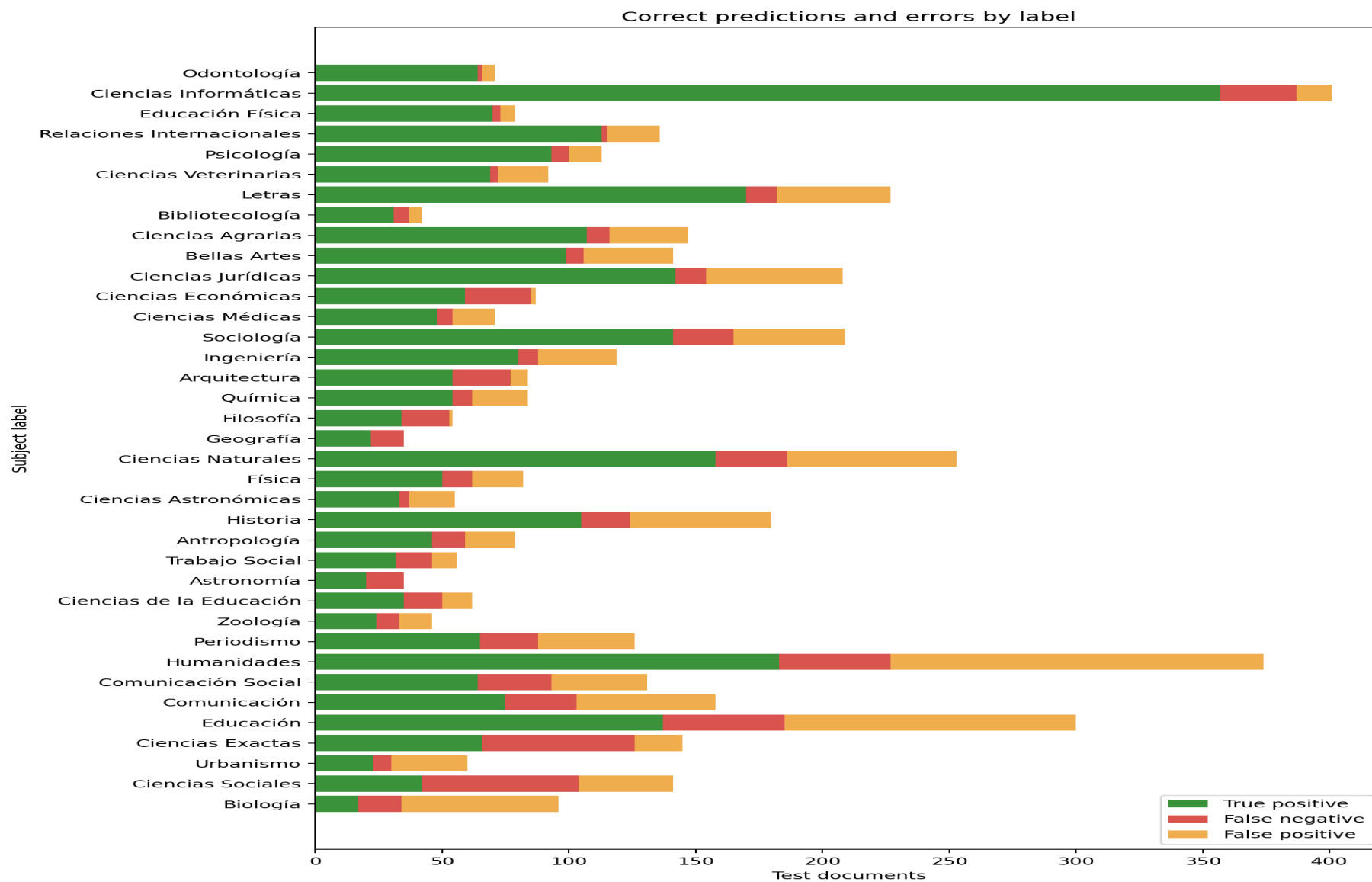


Figure S7. False-positive and false-negative counts.

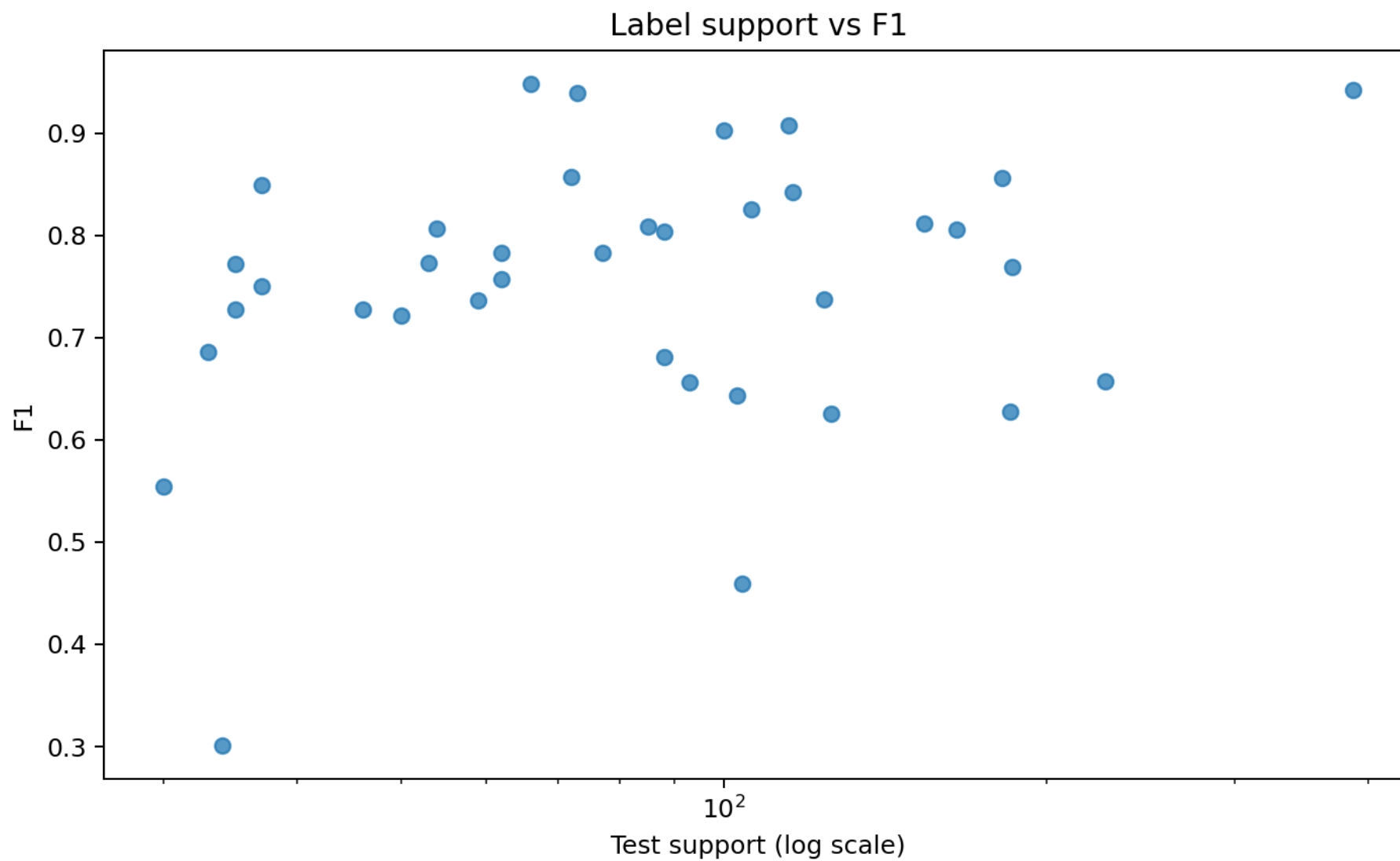


Figure S8. Test support versus subject-level F1.

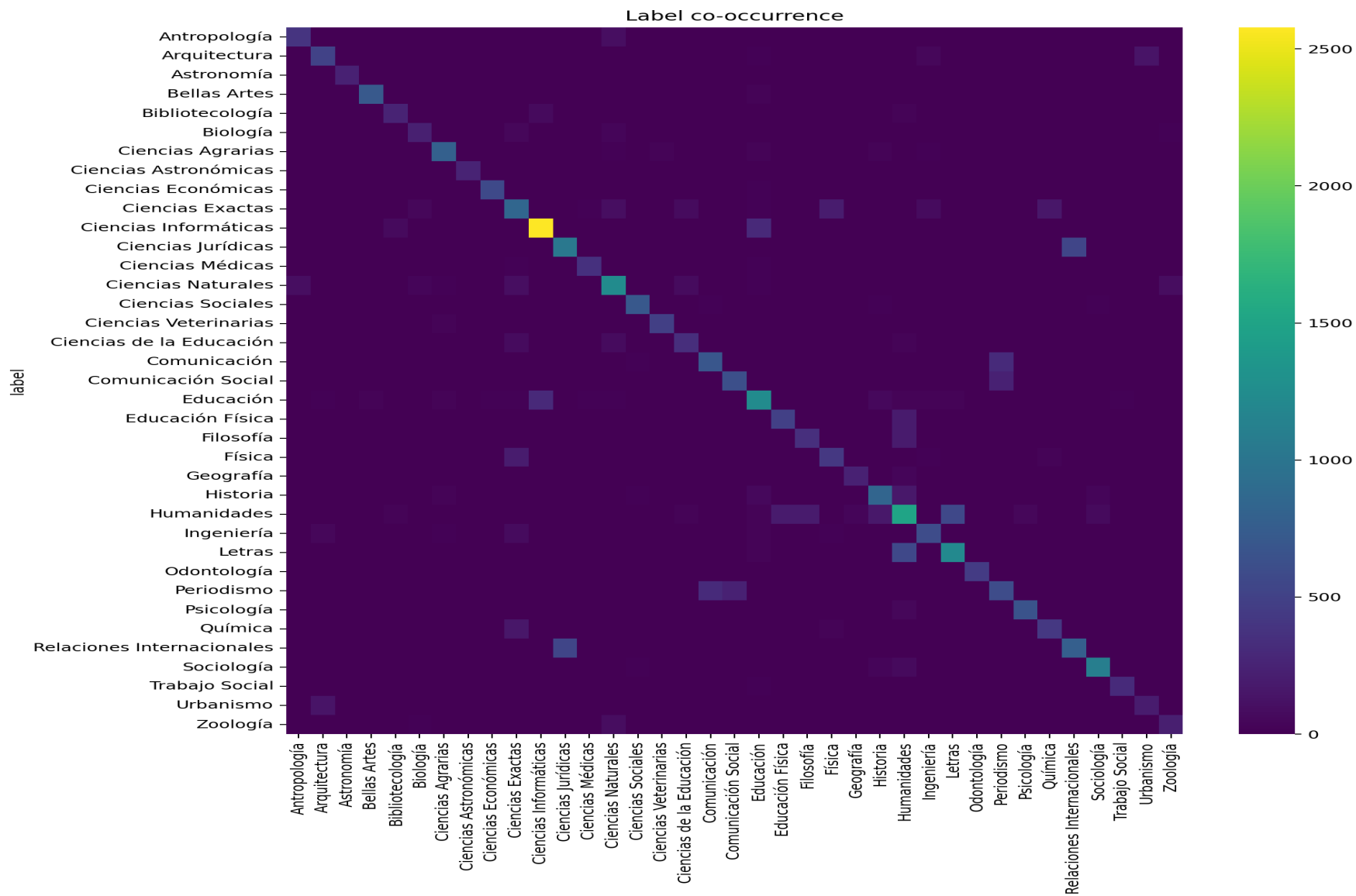


Figure S9. Label co-occurrence structure.

S9. Reproducibility package

Complete aggregate CSV evidence, figures, source configurations, the executable public notebook, and SHA-256 manifest are available at https://github.com/KharolusIII/institutional-repository-subject-classification/tree/main/paper_artifacts